FINANCIAL RISK ANALYTICS BUSINESS REPORT

Guided Project

By

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1. Context

In the realm of modern finance, businesses encounter the perpetual challenge of managing debt obligations effectively to maintain a favourable credit standing and foster sustainable growth. Investors keenly scrutinize companies capable of navigating financial complexities while ensuring stability and profitability. A pivotal instrument in this evaluation process is the balance sheet, which provides a comprehensive overview of a company's assets, liabilities, and shareholder equity, offering insights into its financial health and operational efficiency. In this context, leveraging available financial data, particularly from preceding fiscal periods, becomes imperative for informed decision-making and strategic planning

2. Objective

A renowned credit rating organization wants to develop a Financial Health Assessment Tool. With the help of the tool, it endeavour is to empower businesses and investors with a robust mechanism for evaluating the financial well-being and creditworthiness of companies. By harnessing machine learning techniques, the organization aims to analyse historical financial statements and extract pertinent insights to facilitate informed decision-making via the tool. Specifically, the organization foresees facilitating the following with the help of the tool:

1. Debt Management Analysis: Identify patterns and trends in debt management practices to assess the ability of businesses to fulfil financial obligations promptly and efficiently, and identify potential cases of default.
2. Credit Risk Evaluation: Evaluate credit risk exposure by analysing liquidity ratios, debt-to-equity ratios, and other key financial indicators to ascertain the likelihood of default and inform investment decisions.

As a part of the data science team in the organization, you have been provided with the financial metrics of different companies. The task is to analyse the data provided and develop a predictive model leveraging machine learning techniques to identify whether a given company will default on its debt repayments in the next two quarters. The predictive model will help the organization anticipate potential challenges with the financial performance of the companies and enable proactive risk mitigation strategies.

3. Data Dictionary

The data consists of financial metrics from the balance sheets of different companies. The detailed data dictionary is available in the data dictionary file (FRA_DataDictionary.xlsx).

Sl. No	Column Name	Description
1	Co_Code	Company Code
2	Co_Name	Company Name
3	_Operating_Expense_Rate	Operating Expense Rate: Operating Expenses/Net Sales. The operating expense ratio (OER) is the cost to operate a piece of property compared to the income the property brings in.
4	_Research_and_development_expense_rate	Research and development expense rate: (Research and Development Expenses)/Net Sales. Research and development (R&D) expenses are direct expenditures relating to a company's efforts to develop, design, and enhance its products, services, technologies, or processes.
5	_Cash_flow_rate	Cash flow rate: Cash Flow from Operating/Current Liabilities. Cash flow is a measure of how much cash a business brought in or spent in total over a period of time.
6	_Interest_bearing_debt_interest_rate	Interest-bearing debt interest rate: Interest-bearing Debt/Equity
7	_Tax_rate_A	Tax rate (A): Effective Tax Rate. Effective tax rate represents the percentage of their taxable income that individuals pay in taxes. For corporations, the effective corporate tax rate is the rate they pay on their pre-tax profits.
8	_Cash_Flow_Per_Share	Cash Flow Per Share. It is the after-tax earnings plus depreciation on a per-share basis that functions as a measure of a firm's financial strength
9	_Per_Share_Net_profit_before_tax_Yuan_	Per Share Net profit before tax (Yuan ¥): Pretax Income Per Share. Pretax income, also known as earnings before tax or pretax earnings, is the net income earned by a business before taxes are subtracted/accounted for.
10	_Realized_Sales_Gross_Profit_Growth_Rate	Realized Sales Gross Profit Growth Rate.
11	_Operating_Profit_Growth_Rate	Operating Profit Growth Rate: Operating Income Growth. It is the rate of increase in operating income over the last year.
12	_Continuous_Net_Profit_Growth_Rate	Continuous Net Profit Growth Rate: Net Income-Excluding Disposal Gain or Loss Growth
13	_Total_Asset_Growth_Rate	Total Asset Growth Rate: Total Asset Growth. It is the rate at which how quickly the company has been growing its Assets
14	_Net_Value_Growth_Rate	Net Value Growth Rate: Total Equity Growth
15	_Total_Asset_Return_Growth_Rate_Ratio	Total Asset Return Growth Rate Ratio: Return on Total Asset Growth
16	_Cash_Reinvestment_perc	Cash Reinvestment %: Cash Reinvestment Ratio. It is the valuation ratio that is used to measure the percentage of annual cash flow that the company invests back into the business as a new investment.
17	_Current_Ratio	Current Ratio. The current ratio describes the relationship between a company's assets and liabilities
18	_Quick_Ratio	Quick Ratio: Acid Test. Acid-test ratio (also known as quick ratio) is a measure of a company's liquidity, which is its ability to pay its short-term obligations using only its most liquid assets.
19	_Interest_Expense_Ratio	Interest Expense Ratio: Interest Expenses/Total Revenue
20	_Total_debt_to_Total_net_worth	Total debt/Total net worth: Total Liability/Equity Ratio
21	_Long_term_fund_suitability_ratio_A	Long-term fund suitability ratio (A): (Long-term Liability+Equity)/Fixed Assets
22	_Net_profit_before_tax_to_Paid_in_capital	Net profit before tax/Paid-in capital: Pretax Income/Capital
23	_Total_Asset_Turnover	Total Asset Turnover. Net Sales/Average Total Assets
24	_Accounts_Receivable_Turnover	Accounts Receivable Turnover. The accounts receivable turnover ratio, or receivables turnover, is used in business accounting to quantify how well companies are managing the credit that they extend to their customers by evaluating how long it takes to collect the outstanding debt throughout the accounting period.
25	_Average_Collection_Days	Average Collection Days: Days Receivable Outstanding

26	_Inventory_Turnover_Rate_times	Inventory Turnover Rate (times). The inventory turnover ratio is the number of times a company has sold and replenished its inventory over a specific amount of time. The formula can also be used to calculate the number of days it will take to sell the inventory on hand.
27	_Fixed_Assets_Turnover_Frequency	Fixed Assets Turnover Frequency. Fixed Asset Turnover (FAT) is an efficiency ratio that indicates how well or efficiently a business uses fixed assets to generate sales. This ratio divides net sales by net fixed assets, calculated over an annual period.
28	_Net_Worth_Turnover_Rate_times	Net Worth Turnover Rate (times): Equity Turnover. Equity turnover is a ratio that measures the proportion of a company's sales to its stockholders' equity. The intent of the measurement is to determine the efficiency with which management is using equity to generate revenue.
29	_Operating_profit_per_person	Operating profit per person: Operation Income Per Employee
30	_Allocation_rate_per_person	Allocation rate per person: Fixed Assets Per Employee
31	_Quick_Assets_to_Total_Assets	Quick Assets/Total Assets
32	_Cash_to_Total_Assets	Cash/Total Assets
33	_Quick_Assets_to_Current_Liability	Quick Assets/Current Liability
34	_Cash_to_Current_Liability	Cash/Current Liability
35	_Operating_Funds_to_Liability	Operating Funds to Liability
36	_Inventory_to_Working_Capital	Inventory/Working Capital
37	_Inventory_to_Current_Liability	Inventory/Current Liability
38	_Long_term_Liability_to_Current_Assets	Long-term Liability to Current Assets
39	_Retained_Earnings_to_Total_Assets	Retained Earnings to Total Assets
40	_Total_income_to_Total_expense	Total income/Total expense
41	_Total_expense_to_Assets	Total expense/Assets
42	_Current_Asset_Turnover_Rate	Current Asset Turnover Rate: Current Assets to Sales. The current assets turnover ratio indicates how many times the current assets are turned over in the form of sales within a specific period of time. A higher asset turnover ratio means a better percentage of sales.
43	_Quick_Asset_Turnover_Rate	Quick Asset Turnover Rate: Quick Assets to Sales. The asset turnover ratio measures the efficiency of a company's assets in generating revenue or sales.
44	_Cash_Turnover_Rate	Cash Turnover Rate: Cash to Sales. The cash turnover ratio is an efficiency ratio that reveals the number of times that cash is turned over in an accounting period.
45	_Fixed_Assets_to_Assets	Fixed Assets to Assets. Fixed assets are also known as non-current assets—assets that can't be easily converted into cash.
46	_Cash_Flow_to_Total_Assets	Cash Flow to Total Assets. This ratio indicates the cash a company can generate in relation to its size.
47	_Cash_Flow_to_Liability	Cash Flow to Liability. The amount of money available to run business operations and complete transactions. This is calculated as current assets (cash or near-cash assets, like notes receivable) minus current liabilities (liabilities due during the upcoming accounting period)
48	_CFO_to_Assets	CFO to Assets. Cash flow on total assets is an efficiency ratio that rates cash flows to the company assets without being affected by income recognition or income measurements.
49	_Cash_Flow_to_Equity	Cash Flow to Equity. cash flow to equity is a measure of how much cash is available to the equity shareholders of a company after all expenses, reinvestment, and debt are paid.
50	_Current_Liability_to_Current_Assets	Current Liability to Current Assets. Current liabilities are a company's financial commitments that are due and payable within a year, Current assets are projected to be consumed, sold, or converted into cash within a year or within the operational cycle.
51	_Liability_Assets_Flag	Liability-Assets Flag: 1 if Total Liability exceeds Total Assets, 0 otherwise

52	_Total_assets_to_GNP_price	Total assets to GNP price. Gross National Product (GNP) is the total value of all finished goods and services produced by a country's citizens in a given financial year, irrespective of their location.
53	_No_credit_Interval	No-credit Interval
54	_Degree_of_Financial_Leverage_DFL	Degree of Financial Leverage (DFL). The degree of financial leverage is a financial ratio that measures the sensitivity in fluctuations of a company's overall profitability to the volatility of its operating income caused by changes in its capital structure.
55	_Interest_Coverage_Ratio_Interest_expense_to_EBIT	Interest Coverage Ratio (Interest expense to EBIT). The interest coverage ratio is a debt and profitability ratio used to determine how easily a company can pay interest on its outstanding debt. The interest coverage ratio is calculated by dividing a company's earnings before interest and taxes (EBIT) by its interest expense during a given period.
56	_Net_Income_Flag	Net Income Flag: 1 if Net Income is Negative for the last two years, 0 otherwise
57	_Equity_to_Liability	Equity to Liability Ratio.
58	Default	Whether the Company has Default (Bankrupted) or not? 1 - Defaulted, 0 - Not Defaulted.

4. Data Overview through Python

	Co_Code	Co_Name	_Operating_Expense_Rate	_Research_and_development_expense_rate	_Cash_flow_rate	_Interest_bearing_debt_interest_rate	_Tax_rate_A	_Cash_Flov
0	16974	Hind.Cables	8.820000e+09	0.000000e+00	0.462045	0.000352	0.001417	
1	21214	Tata Tele. Mah.	9.380000e+09	4.230000e+09	0.460116	0.000716	0.000000	
2	14852	ABG Shipyard	3.800000e+09	8.150000e+08	0.449893	0.000496	0.000000	
3	2439	GTL	6.440000e+09	0.000000e+00	0.462731	0.000592	0.009313	
4	23505	Bharati Defence	3.680000e+09	0.000000e+00	0.463117	0.000782	0.400243	

Figure 1: Data head

	Co_Code	Co_Name	_Operating_Expense_Rate	_Research_and_development_expense_rate	_Cash_flow_rate	_Interest_bearing_debt_interest_rate	_Tax_rate_A	_Cash_
2053	2743	Kothari Ferment.	3.021580e-04	6.490000e+09	0.477066	0.000000	0.183014	
2054	21216	Firstobj.Tech.	1.371450e-04	0.000000e+00	0.465211	0.000658	0.000000	
2055	142	Diamines & Chem.	2.114990e-04	8.370000e+09	0.480248	0.000502	0.000000	
2056	18014	IL&FS Engg.	3.750000e+09	0.000000e+00	0.474670	0.000578	0.306205	
2057	43229	Channel Nine	2.981110e-04	0.000000e+00	0.467203	0.000826	0.000000	

Figure 2: Data Tail

This dataset has 2058 rows and 58 columns

```
(2058, 58)
```

Figure 3: Data Shape

5. Checking Datatypes of attributes

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2058 entries, 0 to 2057
Data columns (total 58 columns):
 #   Column                                                Non-Null Count  Dtype
---  ------                                                --------------  -----
 0   Co_Code                                               2058 non-null   int64
 1   Co_Name                                               2058 non-null   object
 2   Operating_Expense_Rate                                2058 non-null   float64
 3   Research_and_development_expense_rate                 2058 non-null   float64
 4   Cash_flow_rate                                        2058 non-null   float64
 5   Interest_bearing_debt_interest_rate                   2058 non-null   float64
 6   Tax_rate_A                                            2058 non-null   float64
 7   Cash_Flow_Per_Share                                   1891 non-null   float64
 8   Per_Share_Net_profit_before_tax_Yuan_                 2058 non-null   float64
 9   Realized_Sales_Gross_Profit_Growth_Rate               2058 non-null   float64
 10  Operating_Profit_Growth_Rate                          2058 non-null   float64
 11  Continuous_Net_Profit_Growth_Rate                     2058 non-null   float64
 12  Total_Asset_Growth_Rate                               2058 non-null   float64
 13  Net_Value_Growth_Rate                                 2058 non-null   float64
 14  Total_Asset_Return_Growth_Rate_Ratio                  2058 non-null   float64
 15  Cash_Reinvestment_perc                                2058 non-null   float64
 16  Current_Ratio                                         2058 non-null   float64
 17  Quick_Ratio                                           2058 non-null   float64
 18  Interest_Expense_Ratio                                2058 non-null   float64
 19  Total_debt_to_Total_net_worth                         2037 non-null   float64
 20  Long_term_fund_suitability_ratio_A                    2058 non-null   float64
 21  Net_profit_before_tax_to_Paid_in_capital              2058 non-null   float64
 22  Total_Asset_Turnover                                  2058 non-null   float64
 23  Accounts_Receivable_Turnover                          2058 non-null   float64
 24  Average_Collection_Days                               2058 non-null   float64
 25  Inventory_Turnover_Rate_times                         2058 non-null   float64
 26  Fixed_Assets_Turnover_Frequency                       2058 non-null   float64
 27  Net_Worth_Turnover_Rate_times                         2058 non-null   float64
 28  Operating_profit_per_person                           2058 non-null   float64
 29  Allocation_rate_per_person                            2058 non-null   float64
 30  Quick_Assets_to_Total_Assets                          2058 non-null   float64
 31  Cash_to_Total_Assets                                  1962 non-null   float64
 32  Quick_Assets_to_Current_Liability                     2058 non-null   float64
 33  Cash_to_Current_Liability                             2058 non-null   float64
 34  Operating_Funds_to_Liability                          2058 non-null   float64
 35  Inventory_to_Working_Capital                          2058 non-null   float64
 36  Inventory_to_Current_Liability                        2058 non-null   float64
 37  Long_term_Liability_to_Current_Assets                 2058 non-null   float64
 38  Retained_Earnings_to_Total_Assets                     2058 non-null   float64
 39  Total_income_to_Total_expense                         2058 non-null   float64
 40  Total_expense_to_Assets                               2058 non-null   float64
 41  Current_Asset_Turnover_Rate                           2058 non-null   float64
 42  Quick_Asset_Turnover_Rate                             2058 non-null   float64
 43  Cash_Turnover_Rate                                    2058 non-null   float64
 44  Fixed_Assets_to_Assets                                2058 non-null   float64
 45  Cash_Flow_to_Total_Assets                             2058 non-null   float64
 46  Cash_Flow_to_Liability                                2058 non-null   float64
 47  CFO_to_Assets                                         2058 non-null   float64
 48  Cash_Flow_to_Equity                                   2058 non-null   float64
 49  Current_Liability_to_Current_Assets                   2044 non-null   float64
 50  Liability_Assets_Flag                                 2058 non-null   int64
 51  Total_assets_to_GNP_price                             2058 non-null   float64
 52  No_credit_Interval                                    2058 non-null   float64
 53  Degree_of_Financial_Leverage_DFL                      2058 non-null   float64
 54  Interest_Coverage_Ratio_Interest_expense_to_EBIT      2058 non-null   float64
 55  Net_Income_Flag                                       2058 non-null   int64
 56  Equity_to_Liability                                   2058 non-null   float64
 57  Default                                               2058 non-null   int64
dtypes: float64(53), int64(4), object(1)
memory usage: 932.7+ KB
```

Figure 4: Data Types

6. Checking for missing values in the data

```
Operating_Expense_Rate                                0
Research_and_development_expense_rate                 0
Cash_flow_rate                                        0
Interest_bearing_debt_interest_rate                   0
Tax_rate_A                                            0
Cash_Flow_Per_Share                                 126
Per_Share_Net_profit_before_tax_Yuan_                 0
Realized_Sales_Gross_Profit_Growth_Rate               0
Operating_Profit_Growth_Rate                          0
Continuous_Net_Profit_Growth_Rate                     0
Total_Asset_Growth_Rate                               0
Net_Value_Growth_Rate                                 0
Total_Asset_Return_Growth_Rate_Ratio                  0
Cash_Reinvestment_perc                                0
Current_Ratio                                         0
Quick_Ratio                                           0
Interest_Expense_Ratio                                0
Total_debt_to_Total_net_worth                        18
Long_term_fund_suitability_ratio_A                    0
Net_profit_before_tax_to_Paid_in_capital              0
Total_Asset_Turnover                                  0
Accounts_Receivable_Turnover                          0
Average_Collection_Days                               0
Inventory_Turnover_Rate_times                         0
Fixed_Assets_Turnover_Frequency                       0
Net_Worth_Turnover_Rate_times                         0
Operating_profit_per_person                           0
Allocation_rate_per_person                            0
Quick_Assets_to_Total_Assets                          0
Cash_to_Total_Assets                                 71
Quick_Assets_to_Current_Liability                     0
Cash_to_Current_Liability                             0
Operating_Funds_to_Liability                          0
Inventory_to_Working_Capital                          0
Inventory_to_Current_Liability                        0
Long_term_Liability_to_Current_Assets                 0
Retained_Earnings_to_Total_Assets                     0
Total_income_to_Total_expense                         0
Total_expense_to_Assets                               0
Current_Asset_Turnover_Rate                           0
Quick_Asset_Turnover_Rate                             0
Cash_Turnover_Rate                                    0
Fixed_Assets_to_Assets                                0
Cash_Flow_to_Total_Assets                             0
Cash_Flow_to_Liability                                0
CFO_to_Assets                                         0
Cash_Flow_to_Equity                                   0
Current_Liability_to_Current_Assets                  11
Total_assets_to_GNP_price                             0
No_credit_Interval                                    0
Degree_of_Financial_Leverage_DFL                      0
Interest_Coverage_Ratio_Interest_expense_to_EBIT      0
Equity_to_Liability                                   0
dtype: int64
```

Figure 5: Missing Values in Data

There are missing values in some columns of data will impute those using KNN imputer.

7. Checking for Duplicate values in Data

```
0
```

Figure 6: No. of Duplicate values

There are no duplicate values in the data.

8. Checking the Statistical Summary of data

	count	mean	std	min	25%	50%	75%	max
Co_Code	2058.0	1.757211e+04	2.189289e+04	4.000000	3.674000e+03	6.240000e+03	2.428075e+04	7.249300e+04
Operating_Expense_Rate	2058.0	2.052389e+09	3.252624e+09	0.000100	1.578727e-04	3.330330e-04	4.110000e+09	9.980000e+09
Research_and_development_expense_rate	2058.0	1.208634e+09	2.144568e+09	0.000000	0.000000e+00	1.994130e-04	1.550000e+09	9.980000e+09
Cash_flow_rate	2058.0	4.652426e-01	2.266269e-02	0.000000	4.600991e-01	4.634450e-01	4.680691e-01	1.000000e+00
Interest_bearing_debt_interest_rate	2058.0	1.113022e+07	9.042595e+07	0.000000	2.760280e-04	4.540450e-04	6.630660e-04	9.900000e+08
Tax_rate_A	2058.0	1.147770e-01	1.524457e-01	0.000000	0.000000e+00	3.709890e-02	2.161909e-01	9.996963e-01
Cash_Flow_Per_Share	1891.0	3.199856e-01	1.529979e-02	0.169449	3.149890e-01	3.206479e-01	3.259178e-01	4.622268e-01
Per_Share_Net_profit_before_tax_Yuan_	2058.0	1.769673e-01	3.015730e-02	0.000000	1.666039e-01	1.756421e-01	1.858854e-01	7.923477e-01
Realized_Sales_Gross_Profit_Growth_Rate	2058.0	2.276117e-02	2.170104e-02	0.004282	2.205831e-02	2.210001e-02	2.215200e-02	1.000000e+00
Operating_Profit_Growth_Rate	2058.0	8.481083e-01	4.589093e-03	0.736430	8.479740e-01	8.480386e-01	8.481147e-01	1.000000e+00
Continuous_Net_Profit_Growth_Rate	2058.0	2.173915e-01	5.678779e-03	0.000000	2.175741e-01	2.175961e-01	2.176198e-01	2.332046e-01
Total_Asset_Growth_Rate	2058.0	5.287663e+09	2.912615e+09	0.000000	4.315000e+09	6.225000e+09	7.220000e+09	9.980000e+09
Net_Value_Growth_Rate	2058.0	5.189504e+06	2.077918e+08	0.000000	4.362833e-04	4.554170e-04	4.883758e-04	9.330000e+09
Total_Asset_Return_Growth_Rate_Ratio	2058.0	2.641004e-01	2.415661e-03	0.251620	2.637383e-01	2.640161e-01	2.643097e-01	3.586288e-01
Cash_Reinvestment_perc	2058.0	3.771970e-01	2.737311e-02	0.025828	3.707295e-01	3.789678e-01	3.855575e-01	1.000000e+00
Current_Ratio	2058.0	1.336249e+06	6.061917e+07	0.000000	6.567062e-03	8.945370e-03	1.350542e-02	2.750000e+09
Quick_Ratio	2058.0	2.775510e+07	4.448654e+08	0.000000	2.946399e-03	5.284241e-03	8.902983e-03	9.230000e+09
Interest_Expense_Ratio	2058.0	6.312913e-01	6.785512e-03	0.525126	6.306116e-01	6.307999e-01	6.317437e-01	8.121652e-01
Total_debt_to_Total_net_worth	2037.0	1.071429e+07	2.696960e+08	0.000000	3.924894e-03	7.270721e-03	1.306869e-02	9.940000e+09
Long_term_fund_suitability_ratio_A	2058.0	8.973310e-03	3.485186e-02	0.004129	5.162031e-03	5.517000e-03	6.415301e-03	1.000000e+00
Net_profit_before_tax_to_Paid_in_capital	2058.0	1.753994e-01	2.622348e-02	0.000000	1.658623e-01	1.745683e-01	1.844450e-01	7.921047e-01
Total_Asset_Turnover	2058.0	1.286405e-01	1.006216e-01	0.000000	6.146927e-02	1.034483e-01	1.679160e-01	9.190405e-01
Accounts_Receivable_Turnover	2058.0	4.159864e+07	5.047673e+08	0.000000	7.446260e-04	1.081432e-03	1.854463e-03	9.740000e+09
Average_Collection_Days	2058.0	2.629786e+07	4.109967e+08	0.000000	3.576384e-03	6.001272e-03	8.638997e-03	8.800000e+09
Inventory_Turnover_Rate_times	2058.0	2.030227e+09	3.077250e+09	0.000000	1.909297e-04	1.910000e+07	3.815000e+09	9.990000e+09
Fixed_Assets_Turnover_Frequency	2058.0	1.230898e+09	2.649289e+09	0.000000	2.278950e-04	5.995245e-04	8.423224e-03	9.990000e+09
Net_Worth_Turnover_Rate_times	2058.0	3.957710e-02	4.239591e-02	0.008871	2.048387e-02	2.870968e-02	4.435484e-02	1.000000e+00
Operating_profit_per_person	2058.0	4.036693e-01	5.358970e-02	0.000000	3.913864e-01	3.950781e-01	4.008927e-01	1.000000e+00
Allocation_rate_per_person	2058.0	5.725559e+06	1.979500e+08	0.000000	4.671612e-03	1.062969e-02	2.457491e-02	8.280000e+09
Quick_Assets_to_Total_Assets	2058.0	3.421979e-01	2.103925e-01	0.000000	1.734827e-01	3.061276e-01	4.845435e-01	9.889440e-01
Cash_to_Total_Assets	1962.0	7.993675e-02	9.862260e-02	0.000184	2.061909e-02	4.563187e-02	9.771301e-02	9.250180e-01
Quick_Assets_to_Current_Liability	2058.0	1.190476e+07	3.122923e+08	0.000000	3.616304e-03	5.972976e-03	9.608533e-03	8.820000e+09
Cash_to_Current_Liability	2058.0	9.282507e+07	7.851899e+08	0.000101	1.085476e-03	2.684338e-03	7.540535e-03	9.170000e+09
Operating_Funds_to_Liability	2058.0	3.482338e-01	3.840302e-02	0.026274	3.377032e-01	3.450257e-01	3.541402e-01	1.000000e+00
Inventory_to_Working_Capital	2058.0	2.777491e-01	1.844394e-02	0.000000	2.770093e-01	2.772511e-01	2.777111e-01	1.000000e+00
Inventory_to_Current_Liability	2058.0	5.786346e+07	6.278795e+08	0.000000	2.890842e-03	6.781166e-03	1.275116e-02	9.600000e+09
Long_term_Liability_to_Current_Assets	2058.0	7.340107e+07	6.693526e+08	0.000000	0.000000e+00	2.587130e-03	1.049684e-02	9.310000e+09
Retained_Earnings_to_Total_Assets	2058.0	9.303546e-01	2.976067e-02	0.000000	9.278868e-01	9.350756e-01	9.409371e-01	9.727688e-01
Total_income_to_Total_expense	2058.0	2.357977e-03	4.644258e-04	0.000000	2.186964e-03	2.297452e-03	2.433146e-03	1.028413e-02
Total_expense_to_Assets	2058.0	3.109208e-02	3.870042e-02	0.000853	1.270426e-02	2.086322e-02	3.530120e-02	1.000000e+00
Current_Asset_Turnover_Rate	2058.0	1.273303e+09	2.839741e+09	0.000000	1.504698e-04	2.461660e-04	1.264005e-03	9.990000e+09
Quick_Asset_Turnover_Rate	2058.0	2.571768e+09	3.453544e+09	0.000000	1.511758e-04	3.794085e-04	5.790000e+09	1.000000e+10
Cash_Turnover_Rate	2058.0	2.653696e+09	2.821245e+09	0.000100	1.737418e-03	1.730000e+09	4.550000e+09	9.990000e+09
Fixed_Assets_to_Assets	2058.0	4.042760e+06	1.834006e+08	0.000000	9.650577e-02	2.138107e-01	4.150287e-01	8.320000e+09
Cash_Flow_to_Total_Assets	2058.0	6.442325e-01	4.505929e-02	0.000000	6.333645e-01	6.432462e-01	6.541577e-01	1.000000e+00
Cash_Flow_to_Liability	2058.0	4.599747e-01	3.288112e-02	0.032583	4.574802e-01	4.593408e-01	4.617433e-01	9.051198e-01
CFO_to_Assets	2058.0	5.797344e-01	6.375060e-02	0.000000	5.503790e-01	5.825431e-01	6.123215e-01	9.751973e-01
Cash_Flow_to_Equity	2058.0	3.146292e-01	1.277967e-02	0.000000	3.127830e-01	3.146423e-01	3.165460e-01	5.692307e-01
Current_Liability_to_Current_Assets	2044.0	3.935178e-02	4.797815e-02	0.000000	2.177539e-02	3.265229e-02	4.394684e-02	1.000000e+00
Liability_Assets_Flag	2058.0	3.401361e-03	5.823606e-02	0.000000	0.000000e+00	0.000000e+00	0.000000e+00	1.000000e+00
Total_assets_to_GNP_price	2058.0	2.779397e+07	4.717714e+08	0.000000	9.124052e-04	2.479550e-03	7.004449e-03	9.820000e+09
No_credit_Interval	2058.0	6.236856e-01	1.163052e-02	0.408682	6.233274e-01	6.237496e-01	6.240452e-01	9.563871e-01
Degree_of_Financial_Leverage_DFL	2058.0	2.785248e-02	1.383854e-02	0.012845	2.677558e-02	2.681466e-02	2.702943e-02	4.643880e-01
Interest_Coverage_Ratio_Interest_expense_to_EBIT	2058.0	5.654355e-01	1.153538e-02	0.172065	5.651580e-01	5.653149e-01	5.662324e-01	6.667613e-01
Net_Income_Flag	2058.0	1.000000e+00	0.000000e+00	1.000000	1.000000e+00	1.000000e+00	1.000000e+00	1.000000e+00
Equity_to_Liability	2058.0	4.252852e-02	5.952518e-02	0.003946	2.040787e-02	2.846004e-02	4.343255e-02	1.000000e+00
Default	2058.0	1.068999e-01	3.090610e-01	0.000000	0.000000e+00	0.000000e+00	0.000000e+00	1.000000e+00

Figure 7: Statistical summary of data

9. Univariate Analysis

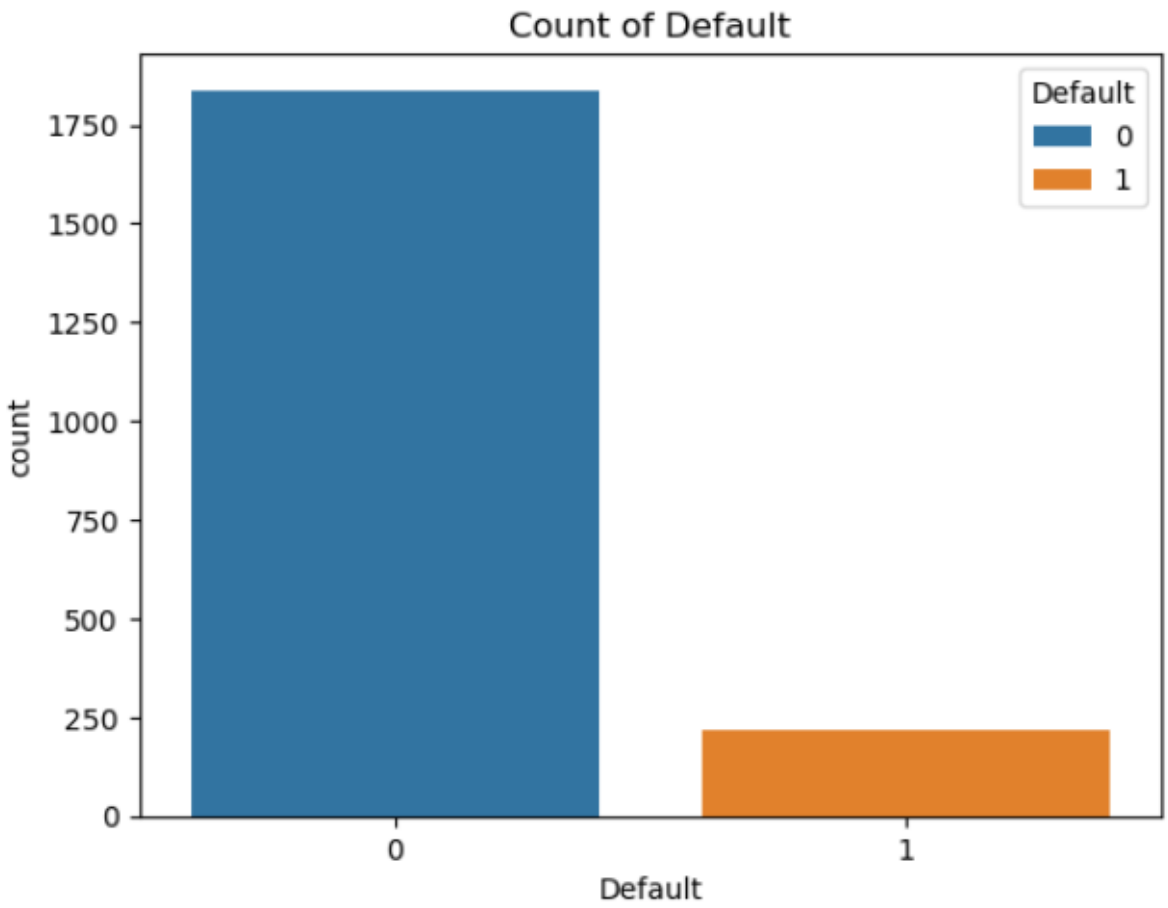

Figure 8: Counts of Target variable

- A large majority of observations (around 1800) fall under Default = 0, meaning no default occurred.
- A much smaller group (around 250) corresponds to Default = 1, indicating defaults
- The dataset is highly imbalanced, with non-defaults significantly outnumbering defaults.
- This imbalance can affect the performance of classification models, especially if accuracy is used as the sole evaluation metric.

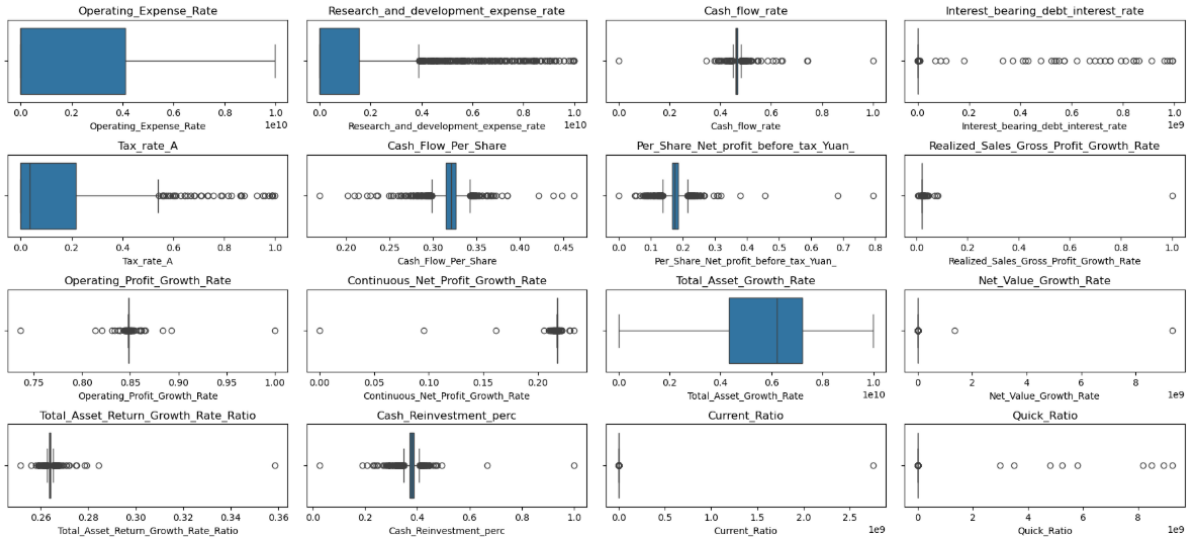

Figure 9: Box plot 1

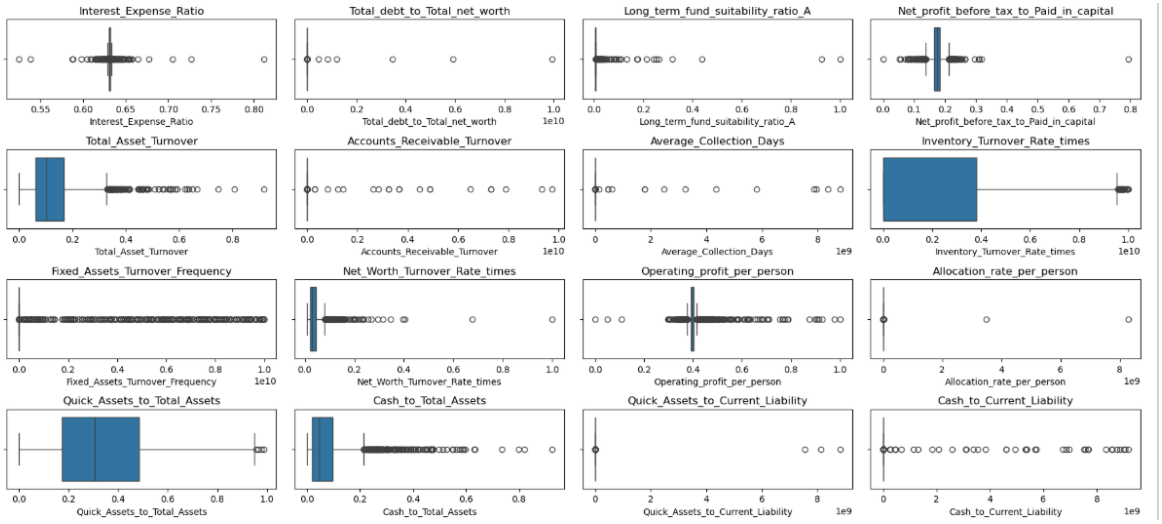

Figure 10: Box plot 2

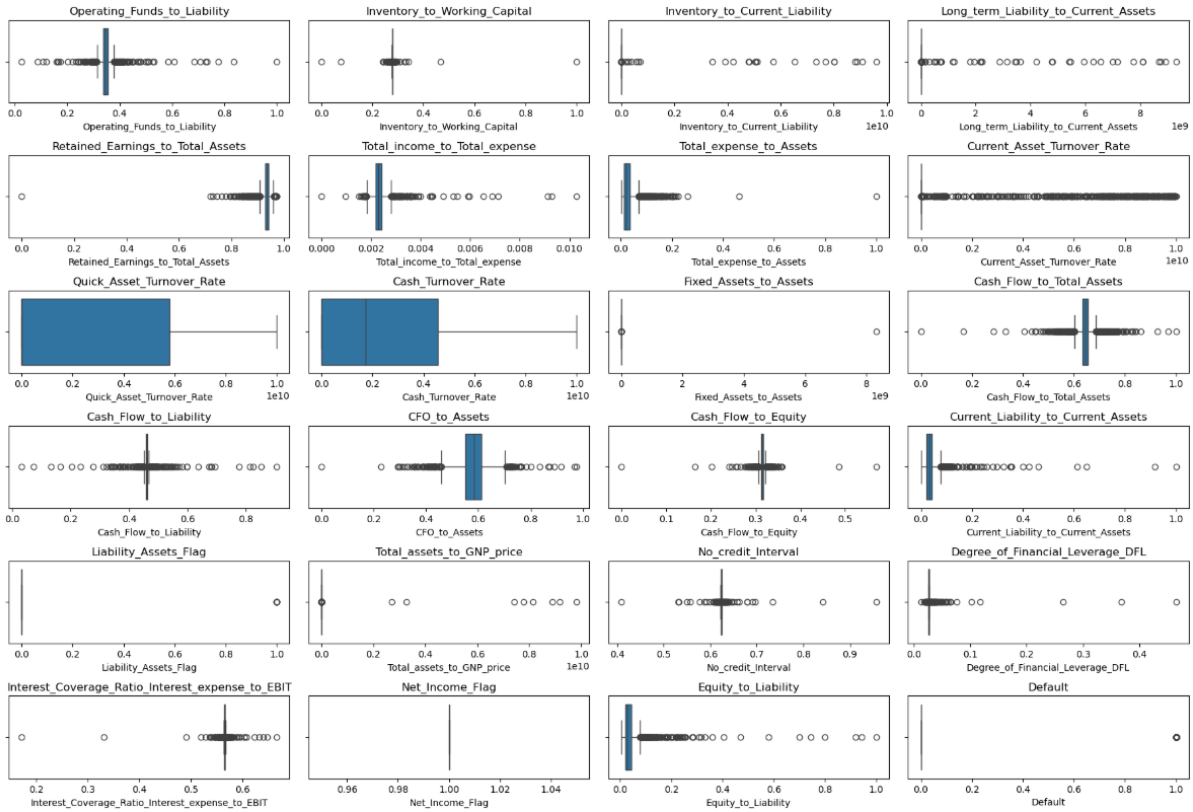

Figure 11: Box plot 3

Observations

- Many financial metrics, such as Cash Flow to Equity, Interest Coverage Ratio, and Inventory to Working Capital, show significant outliers.
- Several variables, including Cash Turnover Rate, Operating Funds to Liability, and CFO to Assets, are right-skewed, indicating that most firms have low values, but a few have exceptionally high ratios.
- Metrics like Net Income Flag, Liability Assets Flag, and Default show minimal variation, with values tightly clustered around 0 or 1.
- Ratios such as Long-Term Liability to Current Assets, Cash Flow to Liability, and Equity to Liability show wide interquartile ranges, indicating substantial variability in financial structure and leverage across firms.
- Metrics like Cash Flow to Total Assets, Interest Coverage Ratio, and Retained Earnings to Total Assets could be strong predictors of default, given their financial relevance and variability.

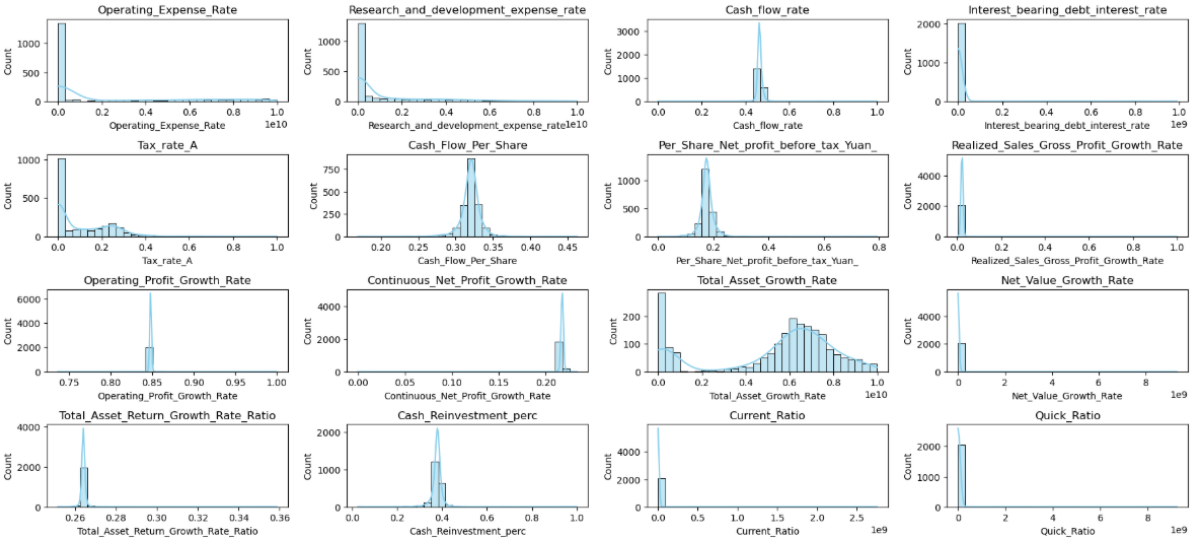

Figure 12: Histogram plot 1

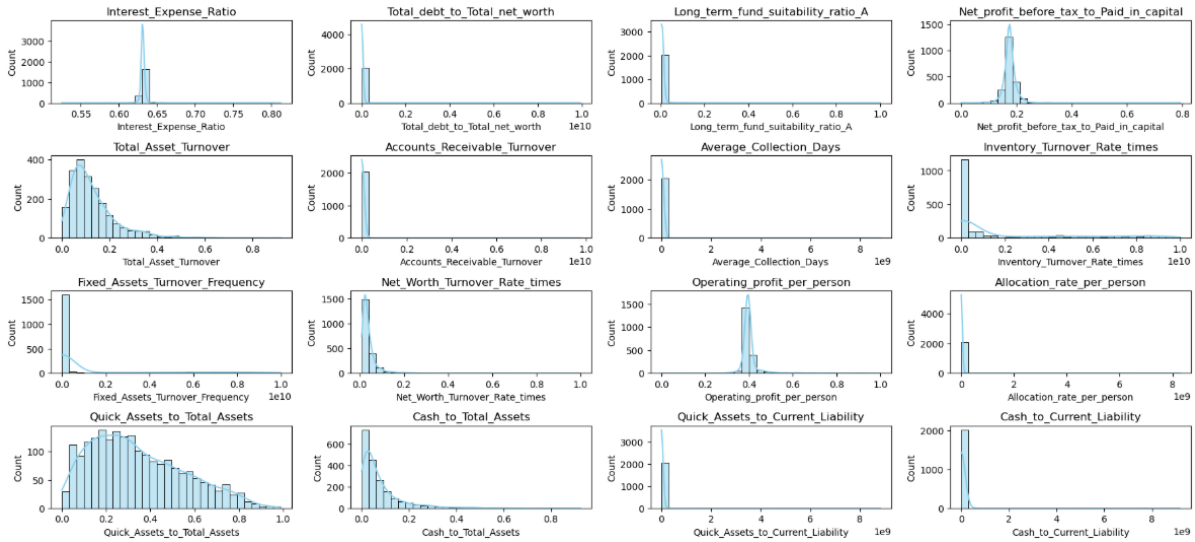

Figure 13: Histogram plot 2

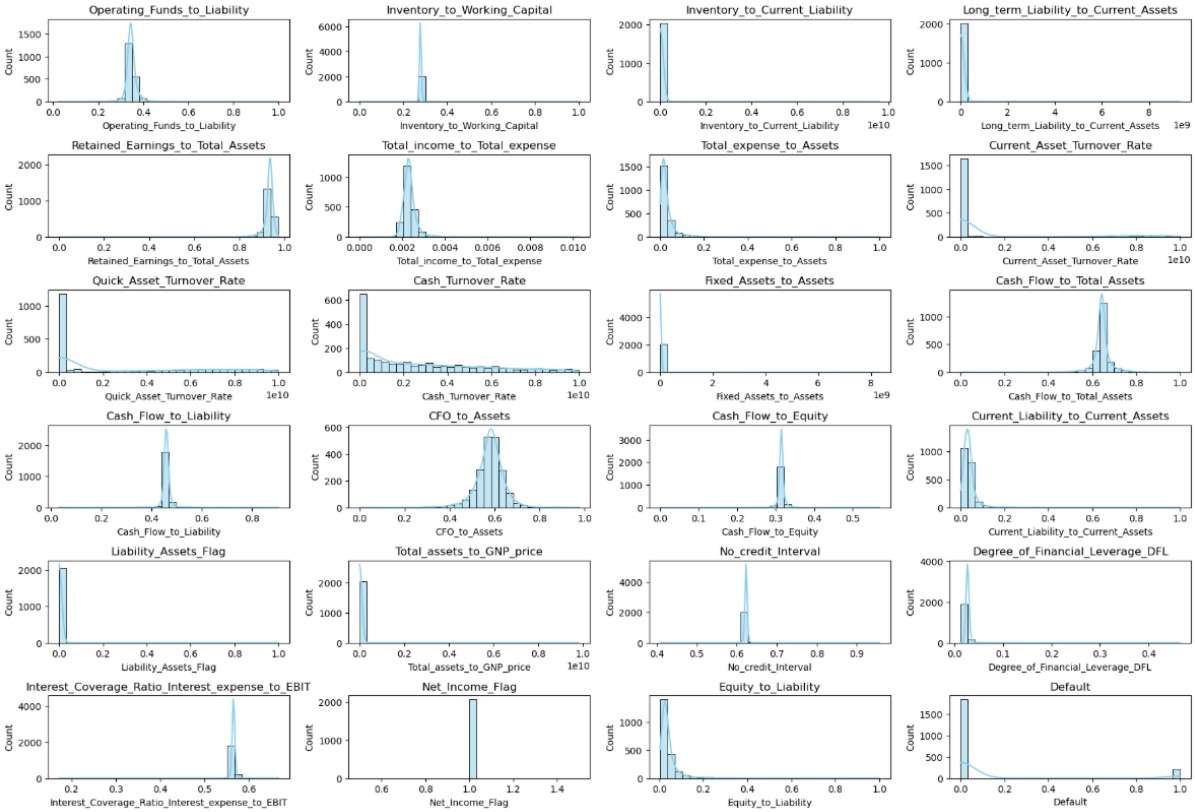

Figure 14: Histogram plot 3

Observations

- Many financial ratios such as Cash Flow to Total Assets, Operating Funds to Liability, and Cash Turnover Rate are right-skewed, meaning most firms have low values, but a few have extremely high ones.
- Variables like Net Income Flag, Liability Assets Flag, and Default show clear binary distributions (0 or 1), confirming their role as categorical indicators.
- Several metrics, including Retained Earnings to Total Assets, Cash Flow to Liability, and CFO to Assets, show a strong concentration near zero.
- Ratios like Long-Term Liability to Current Assets, Equity to Liability, and Interest Coverage Ratio show wide distributions, reflecting diverse capital structures and debt-servicing capacities across firms.
- Metrics such as Current Asset Turnover, Quick Asset Turnover, and Cash Turnover Rate exhibit broad distributions, suggesting significant variation in how efficiently firms utilize their assets

10. Bivariate Analysis

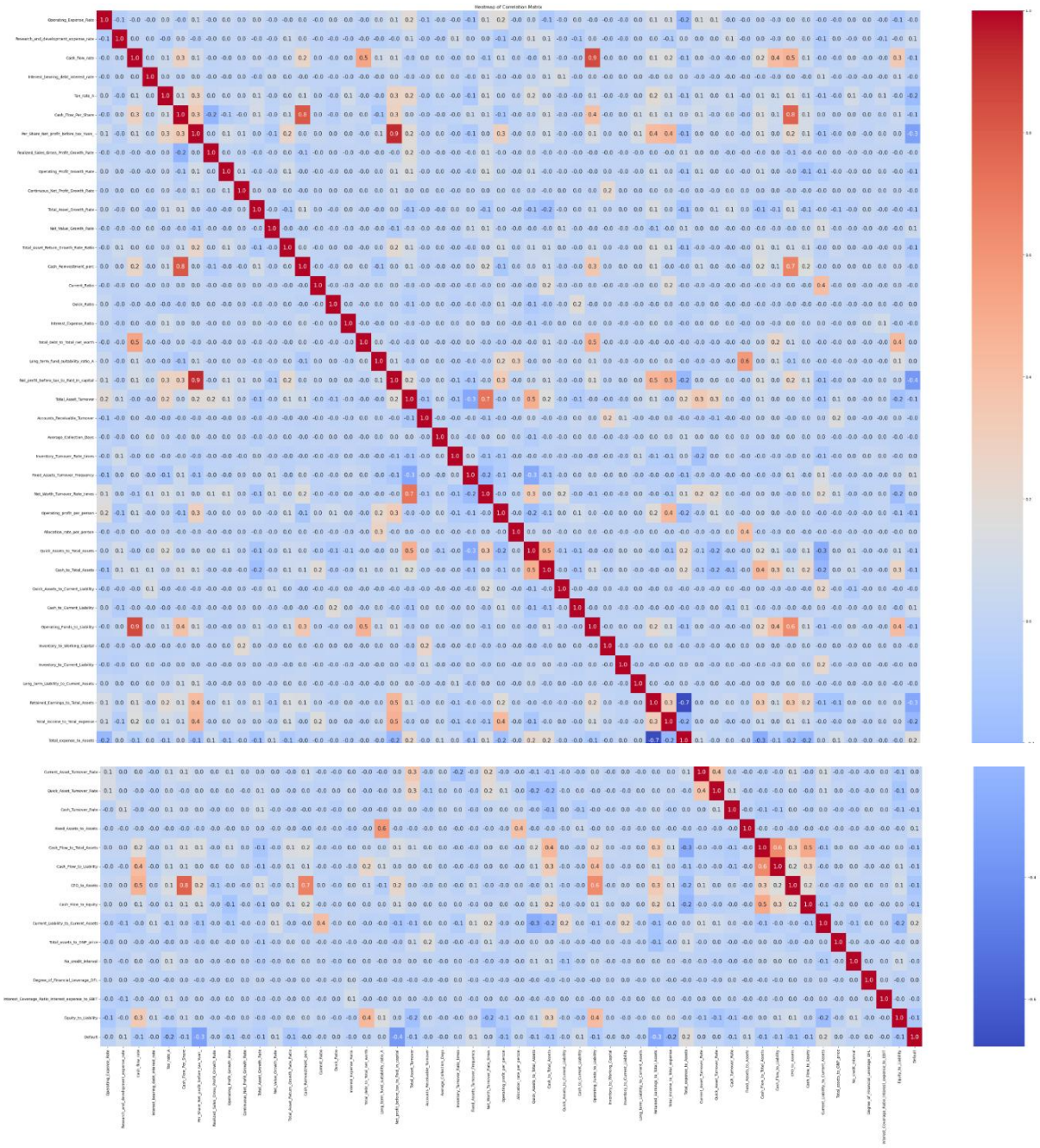

Figure 15: Corelation chart of numerical columns

There are some correlations between variables.

11. Outliers Check

	Column	No. of outliers
0	Operating_Expense_Rate	0
1	Research_and_development_expense_rate	264
2	Cash_flow_rate	206
3	Interest_bearing_debt_interest_rate	94
4	Tax_rate_A	42
5	Cash_Flow_Per_Share	146
6	Per_Share_Net_profit_before_tax_Yuan_	186
7	Realized_Sales_Gross_Profit_Growth_Rate	283
8	Operating_Profit_Growth_Rate	317
9	Continuous_Net_Profit_Growth_Rate	340
10	Total_Asset_Growth_Rate	0
11	Net_Value_Growth_Rate	304
12	Total_Asset_Return_Growth_Rate_Ratio	226
13	Cash_Reinvestment_perc	220
14	Current_Ratio	193
15	Quick_Ratio	190
16	Interest_Expense_Ratio	328
17	Total_debt_to_Total_net_worth	105
18	Long_term_fund_suitability_ratio_A	234
19	Net_profit_before_tax_to_Paid_in_capital	173
20	Total_Asset_Turnover	101
21	Accounts_Receivable_Turnover	281
22	Average_Collection_Days	77
23	Inventory_Turnover_Rate_times	29
24	Fixed_Assets_Turnover_Frequency	501
25	Net_Worth_Turnover_Rate_times	165
26	Operating_profit_per_person	357
27	Allocation_rate_per_person	200
28	Quick_Assets_to_Total_Assets	4
29	Cash_to_Total_Assets	163
30	Quick_Assets_to_Current_Liability	185
31	Cash_to_Current_Liability	253
32	Operating_Funds_to_Liability	219
33	Inventory_to_Working_Capital	247
34	Inventory_to_Current_Liability	129
35	Long_term_Liability_to_Current_Assets	213
36	Retained_Earnings_to_Total_Assets	208
37	Total_income_to_Total_expense	136
38	Total_expense_to_Assets	168
39	Current_Asset_Turnover_Rate	464
40	Quick_Asset_Turnover_Rate	0
41	Cash_Turnover_Rate	0
42	Fixed_Assets_to_Assets	10
43	Cash_Flow_to_Total_Assets	317
44	Cash_Flow_to_Liability	407
45	CFO_to_Assets	110
46	Cash_Flow_to_Equity	306
47	Current_Liability_to_Current_Assets	121
48	Total_assets_to_GNP_price	235
49	No_credit_Interval	396
50	Degree_of_Financial_Leverage_DFL	438
51	Interest_Coverage_Ratio_Interest_expense_to_EBIT	376
52	Equity_to_Liability	190
53	Default	220

Figure 16: Outliers in data

12. Data Preparation

Splitting data into Training and Testing data with 75% and 25% accordingly.

13. Missing values Imputations in Training Data

```
Operating_Expense_Rate                                  0
Research_and_development_expense_rate                   0
Cash_flow_rate                                          0
Interest_bearing_debt_interest_rate                     0
Tax_rate_A                                              0
Cash_Flow_Per_Share                                   126
Per_Share_Net_profit_before_tax_Yuan_                   0
Realized_Sales_Gross_Profit_Growth_Rate                 0
Operating_Profit_Growth_Rate                            0
Continuous_Net_Profit_Growth_Rate                       0
Total_Asset_Growth_Rate                                 0
Net_Value_Growth_Rate                                   0
Total_Asset_Return_Growth_Rate_Ratio                    0
Cash_Reinvestment_perc                                  0
Current_Ratio                                           0
Quick_Ratio                                             0
Interest_Expense_Ratio                                  0
Total_debt_to_Total_net_worth                          18
Long_term_fund_suitability_ratio_A                      0
Net_profit_before_tax_to_Paid_in_capital                0
Total_Asset_Turnover                                    0
Accounts_Receivable_Turnover                            0
Average_Collection_Days                                 0
Inventory_Turnover_Rate_times                           0
Fixed_Assets_Turnover_Frequency                         0
Net_Worth_Turnover_Rate_times                           0
Operating_profit_per_person                             0
Allocation_rate_per_person                              0
Quick_Assets_to_Total_Assets                            0
Cash_to_Total_Assets                                   71
Quick_Assets_to_Current_Liability                       0
Cash_to_Current_Liability                               0
Operating_Funds_to_Liability                            0
Inventory_to_Working_Capital                            0
Inventory_to_Current_Liability                          0
Long_term_Liability_to_Current_Assets                   0
Retained_Earnings_to_Total_Assets                       0
Total_income_to_Total_expense                           0
Total_expense_to_Assets                                 0
Current_Asset_Turnover_Rate                             0
Quick_Asset_Turnover_Rate                               0
Cash_Turnover_Rate                                      0
Fixed_Assets_to_Assets                                  0
Cash_Flow_to_Total_Assets                               0
Cash_Flow_to_Liability                                  0
CFO_to_Assets                                           0
Cash_Flow_to_Equity                                     0
Current_Liability_to_Current_Assets                    11
Total_assets_to_GNP_price                               0
No_credit_Interval                                      0
Degree_of_Financial_Leverage_DFL                        0
Interest_Coverage_Ratio_Interest_expense_to_EBIT        0
Equity_to_Liability                                     0
dtype: int64
```

Figure 17: Missing values in training data

14. Missing values Imputations in Test Data

```
Operating_Expense_Rate                                0
Research_and_development_expense_rate                 0
Cash_flow_rate                                        0
Interest_bearing_debt_interest_rate                   0
Tax_rate_A                                            0
Cash_Flow_Per_Share                                  41
Per_Share_Net_profit_before_tax_Yuan_                 0
Realized_Sales_Gross_Profit_Growth_Rate               0
Operating_Profit_Growth_Rate                          0
Continuous_Net_Profit_Growth_Rate                     0
Total_Asset_Growth_Rate                               0
Net_Value_Growth_Rate                                 0
Total_Asset_Return_Growth_Rate_Ratio                  0
Cash_Reinvestment_perc                                0
Current_Ratio                                         0
Quick_Ratio                                           0
Interest_Expense_Ratio                                0
Total_debt_to_Total_net_worth                         3
Long_term_fund_suitability_ratio_A                    0
Net_profit_before_tax_to_Paid_in_capital              0
Total_Asset_Turnover                                  0
Accounts_Receivable_Turnover                          0
Average_Collection_Days                               0
Inventory_Turnover_Rate_times                         0
Fixed_Assets_Turnover_Frequency                       0
Net_Worth_Turnover_Rate_times                         0
Operating_profit_per_person                           0
Allocation_rate_per_person                            0
Quick_Assets_to_Total_Assets                          0
Cash_to_Total_Assets                                 25
Quick_Assets_to_Current_Liability                     0
Cash_to_Current_Liability                             0
Operating_Funds_to_Liability                          0
Inventory_to_Working_Capital                          0
Inventory_to_Current_Liability                        0
Long_term_Liability_to_Current_Assets                 0
Retained_Earnings_to_Total_Assets                     0
Total_income_to_Total_expense                         0
Total_expense_to_Assets                               0
Current_Asset_Turnover_Rate                           0
Quick_Asset_Turnover_Rate                             0
Cash_Turnover_Rate                                    0
Fixed_Assets_to_Assets                                0
Cash_Flow_to_Total_Assets                             0
Cash_Flow_to_Liability                                0
CFO_to_Assets                                         0
Cash_Flow_to_Equity                                   0
Current_Liability_to_Current_Assets                   3
Total_assets_to_GNP_price                             0
No_credit_Interval                                    0
Degree_of_Financial_Leverage_DFL                      0
Interest_Coverage_Ratio_Interest_expense_to_EBIT      0
Equity_to_Liability                                   0
dtype: int64
```

Figure 18: Missing values in test data

15. Scaling Data

	Operating_Expense_Rate	Research_and_development_expense_rate	Cash_flow_rate	Interest_bearing_debt_interest_rate	Tax_rate_A	Cash_Flow_Per_Share	Per_Share_Net_
0	-0.633296	-0.396806	-0.132455	-0.128462	-0.754347	0.088170	
1	-0.633296	-0.561672	-0.934352	-0.128462	-0.754347	-1.224514	
2	-0.633296	0.361946	-0.290335	-0.128462	0.061964	-0.409659	
3	-0.633296	-0.561672	-0.179548	-0.128462	-0.754347	-0.077773	
4	-0.633296	-0.561672	-0.123892	-0.128462	-0.754347	-0.168422	

5 rows × 53 columns

Figure 19: Scaled data Head

	Operating_Expense_Rate	Research_and_development_expense_rate	Cash_flow_rate	Interest_bearing_debt_interest_rate	Tax_rate_A	Cash_Flow_Per_Share	Per_Share_Net_
0	-0.633296	1.539557	0.118477	-0.128462	-0.754347	0.053496	
1	2.016795	0.135659	0.441752	-0.128462	-0.754347	0.164950	
2	-0.633296	0.177222	-0.141279	-0.128462	0.085538	0.112938	
3	-0.633296	2.144527	-0.666470	-0.128462	-0.754347	-0.986745	
4	0.763746	0.301910	-0.050325	-0.128462	0.719997	0.578569	

rows × 53 columns

Figure 20: Scaled Data tail

16. Logistic Regression Model

```
Warning: Maximum number of iterations has been exceeded.
         Current function value: 0.193946
         Iterations: 35
                           Logit Regression Results
==============================================================================
Dep. Variable:                Default   No. Observations:                 1543
Model:                          Logit   Df Residuals:                     1489
Method:                           MLE   Df Model:                           53
Date:                Sun, 06 Jul 2025   Pseudo R-squ.:                  0.4297
Time:                        15:21:14   Log-Likelihood:                -299.26
converged:                       False   LL-Null:                       -524.71
Covariance Type:            nonrobust   LLR p-value:                 1.764e-64
=====================================================================================================================
                                                       coef    std err          z      P>|z|      [0.025      0.975]
---------------------------------------------------------------------------------------------------------------------
const                                             -7.4685   2410.787     -0.003      0.998   -4732.524    4717.587
Operating_Expense_Rate                             0.2077      0.121      1.713      0.087      -0.030       0.445
Research_and_development_expense_rate              0.3556      0.104      3.433      0.001       0.153       0.559
Cash_flow_rate                                    -0.1837      1.016     -0.181      0.857      -2.175       1.808
Interest_bearing_debt_interest_rate                0.1755      0.151      1.163      0.245      -0.120       0.471
Tax_rate_A                                        -0.2580      0.174     -1.481      0.139      -0.599       0.083
Cash_Flow_Per_Share                               -0.3533      0.281     -1.260      0.208      -0.903       0.196
Per_Share_Net_profit_before_tax_Yuan_              0.2518      1.276      0.197      0.844      -2.249       2.752
Realized_Sales_Gross_Profit_Growth_Rate            0.1012      0.118      0.859      0.390      -0.130       0.332
Operating_Profit_Growth_Rate                      -0.1546      0.267     -0.579      0.563      -0.678       0.369
Continuous_Net_Profit_Growth_Rate                  0.1736      0.132      1.317      0.188      -0.085       0.432
Total_Asset_Growth_Rate                           -0.0640      0.131     -0.487      0.626      -0.321       0.193
Net_Value_Growth_Rate                              0.5177   3097.822      0.000      1.000   -6071.102    6072.138
Total_Asset_Return_Growth_Rate_Ratio              -0.3299      0.361     -0.915      0.360      -1.037       0.377
Cash_Reinvestment_perc                             0.1700      0.346      0.491      0.624      -0.509       0.849
Current_Ratio                                     -1.6114      0.925     -1.742      0.081      -3.424       0.201
Quick_Ratio                                       -2.7355   2.57e+04     -0.000      1.000   -5.05e+04    5.05e+04
Interest_Expense_Ratio                             0.0197      0.065      0.303      0.762      -0.107       0.147
Total_debt_to_Total_net_worth                      1.9035      0.623      3.058      0.002       0.683       3.124
Long_term_fund_suitability_ratio_A                 0.1675      0.223      0.751      0.452      -0.269       0.604
Net_profit_before_tax_to_Paid_in_capital          -1.0834      1.179     -0.919      0.358      -3.394       1.227
Total_Asset_Turnover                              -0.2122      0.319     -0.666      0.506      -0.837       0.413
Accounts_Receivable_Turnover                      -1.0019      0.642     -1.560      0.119      -2.261       0.257
Average_Collection_Days                          -15.1938   2.49e+04     -0.001      1.000   -4.89e+04    4.88e+04
Inventory_Turnover_Rate_times                     -0.0490      0.117     -0.420      0.675      -0.278       0.180
Fixed_Assets_Turnover_Frequency                    0.1775      0.106      1.678      0.093      -0.030       0.385
Net_Worth_Turnover_Rate_times                     -0.2559      0.211     -1.212      0.225      -0.670       0.158
Operating_profit_per_person                        0.0505      0.195      0.259      0.796      -0.331       0.432
Allocation_rate_per_person                       -80.4893    153.634     -0.524      0.600    -381.606     220.627
Quick_Assets_to_Total_Assets                       0.1935      0.189      1.024      0.306      -0.177       0.564
Cash_to_Total_Assets                              -0.3059      0.222     -1.380      0.168      -0.740       0.129
Quick_Assets_to_Current_Liability                 -0.5860   1.49e+04  -3.92e-05      1.000   -2.93e+04    2.93e+04
Cash_to_Current_Liability                          0.0684      0.076      0.905      0.365      -0.080       0.217
Operating_Funds_to_Liability                       1.2409      0.783      1.584      0.113      -0.294       2.776
Inventory_to_Working_Capital                      -0.1714      0.158     -1.088      0.276      -0.480       0.137
Inventory_to_Current_Liability                     0.1022      0.117      0.870      0.384      -0.128       0.332
Long_term_Liability_to_Current_Assets             -0.0208      0.107     -0.195      0.846      -0.230       0.188
Retained_Earnings_to_Total_Assets                 -0.2111      0.207     -1.019      0.308      -0.617       0.195
Total_income_to_Total_expense                     -1.4219      0.437     -3.252      0.001      -2.279      -0.565
Total_expense_to_Assets                            0.0849      0.253      0.335      0.738      -0.412       0.582
Current_Asset_Turnover_Rate                       -0.0962      0.129     -0.746      0.456      -0.349       0.157
Quick_Asset_Turnover_Rate                          0.0640      0.128      0.499      0.618      -0.188       0.316
Cash_Turnover_Rate                                -0.4286      0.130     -3.307      0.001      -0.683      -0.175
Fixed_Assets_to_Assets                            31.5359    195.727      0.161      0.872    -352.082     415.154
Cash_Flow_to_Total_Assets                          0.9901      0.270      3.668      0.000       0.461       1.519
Cash_Flow_to_Liability                            -2.7554      0.607     -4.542      0.000      -3.945      -1.566
CFO_to_Assets                                     -0.3143      0.467     -0.673      0.501      -1.230       0.602
Cash_Flow_to_Equity                               -0.0344      0.085     -0.404      0.686      -0.201       0.132
Current_Liability_to_Current_Assets               -0.0863      0.121     -0.714      0.476      -0.323       0.151
Total_assets_to_GNP_price                         -0.0290      0.076     -0.384      0.701      -0.177       0.119
No_credit_Interval                                 0.1051      0.079      1.326      0.185      -0.050       0.260
Degree_of_Financial_Leverage_DFL                   0.0729      0.056      1.303      0.193      -0.037       0.183
Interest_Coverage_Ratio_Interest_expense_to_EBIT   0.0677      0.087      0.778      0.437      -0.103       0.238
Equity_to_Liability                               -3.0217      0.709     -4.260      0.000      -4.412      -1.632
=====================================================================================================================
```

Figure 21: Logistic Regression summary

17. Logistic Regression Model - Training Performance

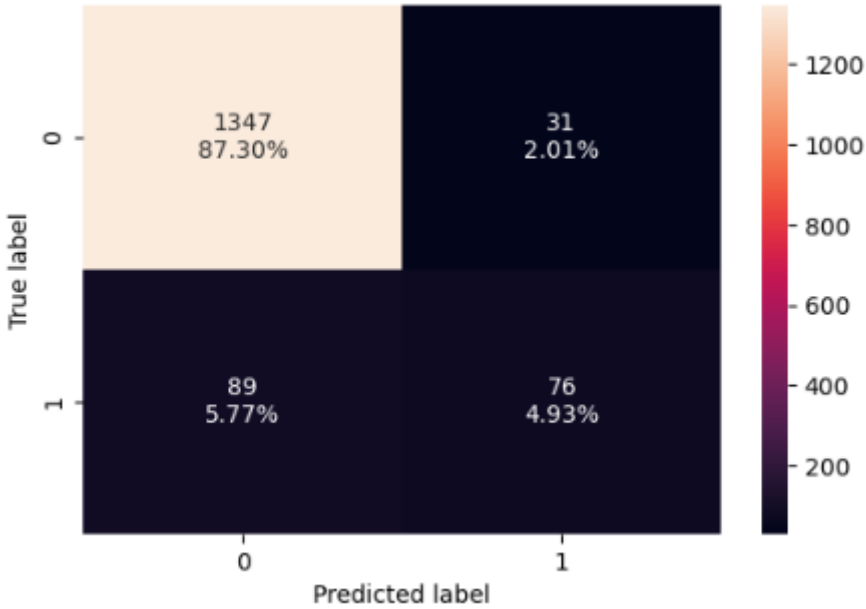

Figure 22: Confusion Matrix of Model

	Accuracy	Recall	Precision	F1
0	0.922229	0.460606	0.71028	0.558824

Figure 23: Model Performance

18. Logistic Regression Model - Test Performance

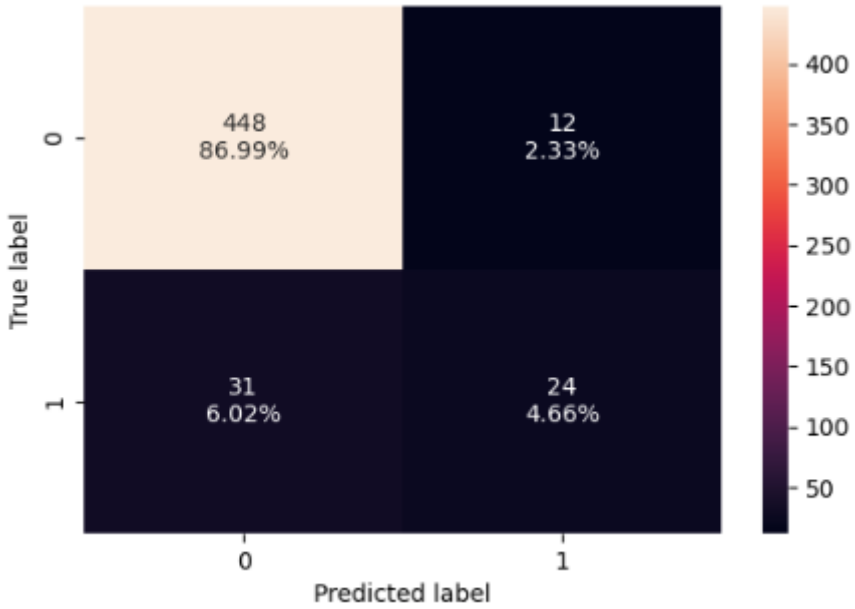

Figure 24: Confusion Matrix of Model

	Accuracy	Recall	Precision	F1
0	0.916505	0.436364	0.666667	0.527473

Figure 25: Model Performance

19. Random Forest Model

20. Random Forest Model - Training Performance

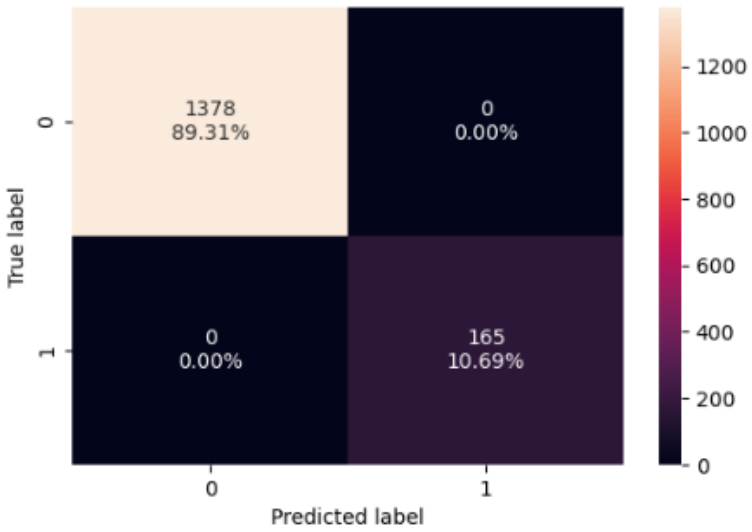

Figure 26: Confusion Matrix of Model

	Accuracy	Recall	Precision	F1
0	1.0	1.0	1.0	1.0

Figure 27: Model Performance

21. Random Forest Model - Test Performance

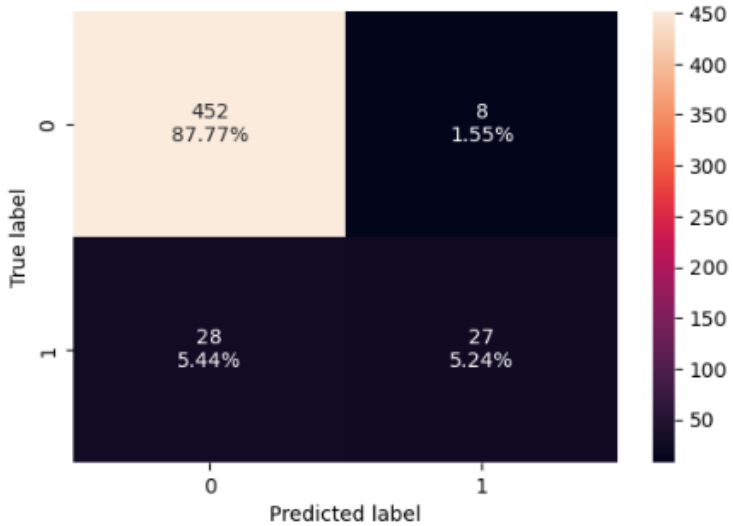

Figure 28: Confusion Matrix of Model

	Accuracy	Recall	Precision	F1
0	0.930097	0.490909	0.771429	0.6

Figure 29: Model Performance

22. Model Performance Improvement - Logistic Regression

Variance Inflation Factors:

	Variable	VIF
0	Operating_Expense_Rate	1.254220
1	Research_and_development_expense_rate	1.444929
2	Cash_flow_rate	6857.275398
3	Interest_bearing_debt_interest_rate	1.049490
4	Tax_rate_A	1.954006
5	Cash_Flow_Per_Share	2239.472240
6	Per_Share_Net_profit_before_tax_Yuan_	284.688016
7	Realized_Sales_Gross_Profit_Growth_Rate	249.698263
8	Operating_Profit_Growth_Rate	0.591271
9	Continuous_Net_Profit_Growth_Rate	1626.146440
10	Total_Asset_Growth_Rate	4.911475
11	Net_Value_Growth_Rate	1.045037
12	Total_Asset_Return_Growth_Rate_Ratio	9466.052035
13	Cash_Reinvestment_perc	2237.299349
14	Current_Ratio	6.392733
15	Quick_Ratio	1.067907
16	Interest_Expense_Ratio	6460.415120
17	Total_debt_to_Total_net_worth	3.718359
18	Long_term_fund_suitability_ratio_A	1.943884
19	Net_profit_before_tax_to_Paid_in_capital	379.455914
20	Total_Asset_Turnover	15.033087
21	Accounts_Receivable_Turnover	1.071284
22	Average_Collection_Days	1.065887
23	Inventory_Turnover_Rate_times	1.100156
24	Fixed_Assets_Turnover_Frequency	1.491294
25	Net_Worth_Turnover_Rate_times	9.354351
26	Operating_profit_per_person	97.197535
27	Allocation_rate_per_person	1.199823
28	Quick_Assets_to_Total_Assets	8.745619
29	Cash_to_Total_Assets	3.610059
30	Quick_Assets_to_Current_Liability	1.010228
31	Cash_to_Current_Liability	1.095831
32	Operating_Funds_to_Liability	1178.296977
33	Inventory_to_Working_Capital	1758.509692
34	Inventory_to_Current_Liability	1.135347
35	Long_term_Liability_to_Current_Assets	1.115536
36	Retained_Earnings_to_Total_Assets	2664.220029
37	Total_income_to_Total_expense	68.248641
38	Total_expense_to_Assets	5.155590
39	Current_Asset_Turnover_Rate	1.697568
40	Quick_Asset_Turnover_Rate	1.376532
41	Cash_Turnover_Rate	1.106589
42	Fixed_Assets_to_Assets	1.816361
43	Cash_Flow_to_Total_Assets	622.306781
44	Cash_Flow_to_Liability	579.290515
45	CFO_to_Assets	917.810175
46	Cash_Flow_to_Equity	685.062567
47	Current_Liability_to_Current_Assets	2.815379
48	Total_assets_to_GNP_price	1.044044
49	No_credit_Interval	2486.675790
50	Degree_of_Financial_Leverage_DFL	7.290176
51	Interest_Coverage_Ratio_Interest_expense_to_EBIT	1835.587699
52	Equity_to_Liability	7.694183

Figure 30: Variance Inflation Factors

```
['Cash_flow_rate',
 'Cash_Flow_Per_Share',
 'Per_Share_Net_profit_before_tax_Yuan_',
 'Realized_Sales_Gross_Profit_Growth_Rate',
 'Continuous_Net_Profit_Growth_Rate',
 'Total_Asset_Return_Growth_Rate_Ratio',
 'Cash_Reinvestment_perc',
 'Current_Ratio',
 'Interest_Expense_Ratio',
 'Net_profit_before_tax_to_Paid_in_capital',
 'Total_Asset_Turnover',
 'Net_Worth_Turnover_Rate_times',
 'Operating_profit_per_person',
 'Quick_Assets_to_Total_Assets',
 'Operating_Funds_to_Liability',
 'Inventory_to_Working_Capital',
 'Retained_Earnings_to_Total_Assets',
 'Total_income_to_Total_expense',
 'Total_expense_to_Assets',
 'Cash_Flow_to_Total_Assets',
 'Cash_Flow_to_Liability',
 'CFO_to_Assets',
 'Cash_Flow_to_Equity',
 'No_credit_Interval',
 'Degree_of_Financial_Leverage_DFL',
 'Interest_Coverage_Ratio_Interest_expense_to_EBIT',
 'Equity_to_Liability']
```

Figure 31: Variables having high VIF

```
(1543, 26)
```

Figure 32: Shape of Training Data

```
(515, 26)
```

Figure 33: Shape of test Data

```
Warning: Maximum number of iterations has been exceeded.
         Current function value: 0.272896
         Iterations: 35
                           Logit Regression Results
==============================================================================
Dep. Variable:                Default   No. Observations:                 1543
Model:                          Logit   Df Residuals:                     1516
Method:                           MLE   Df Model:                           26
Date:                Sun, 06 Jul 2025   Pseudo R-squ.:                  0.1975
Time:                        15:26:47   Log-Likelihood:                -421.08
converged:                       False   LL-Null:                       -524.71
Covariance Type:            nonrobust   LLR p-value:                 3.567e-30
==========================================================================================
                                         coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------------------
const                                 -8.5564    2.9e+06  -2.95e-06      1.000   -5.68e+06    5.68e+06
Operating_Expense_Rate                 0.0230      0.092      0.251      0.802      -0.157       0.203
Research_and_development_expense_rate  0.2876      0.079      3.634      0.000       0.133       0.443
Interest_bearing_debt_interest_rate   -0.0864      0.148     -0.585      0.559      -0.376       0.203
Tax_rate_A                            -1.3551      0.209     -6.486      0.000      -1.765      -0.946
Operating_Profit_Growth_Rate          -1.0005      0.432     -2.317      0.020      -1.847      -0.154
Total_Asset_Growth_Rate               -0.1710      0.096     -1.784      0.074      -0.359       0.017
Net_Value_Growth_Rate                  0.6984   3.34e+04   2.09e-05      1.000   -6.54e+04    6.54e+04
Quick_Ratio                           -3.0354   2.79e+04     -0.000      1.000   -5.47e+04    5.47e+04
Total_debt_to_Total_net_worth          0.0725      0.072      1.002      0.316      -0.069       0.214
Long_term_fund_suitability_ratio_A    -0.0578      0.176     -0.328      0.743      -0.403       0.288
Accounts_Receivable_Turnover          -0.2766      0.257     -1.077      0.282      -0.780       0.227
Average_Collection_Days               -9.3866    273.902     -0.034      0.973    -546.225     527.452
Inventory_Turnover_Rate_times          0.1133      0.092      1.235      0.217      -0.067       0.293
Fixed_Assets_Turnover_Frequency        0.1207      0.090      1.344      0.179      -0.055       0.297
Allocation_rate_per_person          -208.2395   9.98e+05     -0.000      1.000   -1.96e+06    1.96e+06
Cash_to_Total_Assets                  -0.4939      0.152     -3.241      0.001      -0.793      -0.195
Quick_Assets_to_Current_Liability     -0.5620   3580.042     -0.000      1.000   -7017.315    7016.191
Cash_to_Current_Liability              0.1466      0.068      2.160      0.031       0.014       0.280
Inventory_to_Current_Liability        -0.0842      0.110     -0.762      0.446      -0.300       0.132
Long_term_Liability_to_Current_Assets  0.0459      0.091      0.506      0.613      -0.132       0.224
Current_Asset_Turnover_Rate           -0.0281      0.103     -0.273      0.785      -0.230       0.174
Quick_Asset_Turnover_Rate             -0.0296      0.100     -0.297      0.766      -0.225       0.166
Cash_Turnover_Rate                    -0.3095      0.106     -2.933      0.003      -0.516      -0.103
Fixed_Assets_to_Assets                83.7053   1.14e+08   7.36e-07      1.000   -2.23e+08    2.23e+08
Current_Liability_to_Current_Assets    0.3092      0.093      3.336      0.001       0.128       0.491
Total_assets_to_GNP_price              0.0502      0.058      0.859      0.391      -0.064       0.165
==========================================================================================
```

Figure 34: Tuned Logistic Regression Model Summary

Optimal Threshold Value: 0.117

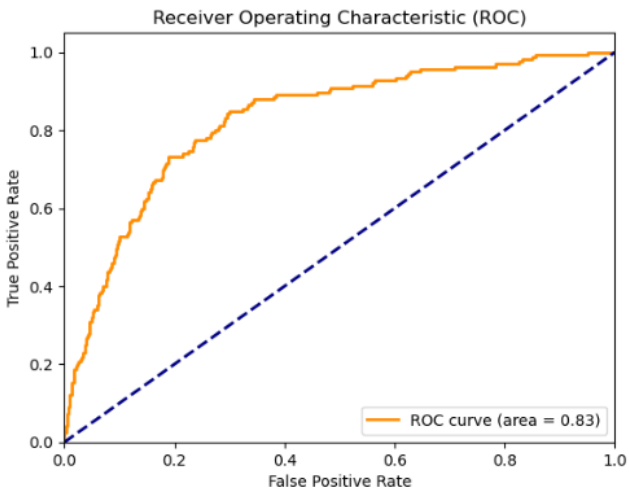

Figure 35: ROC Curve

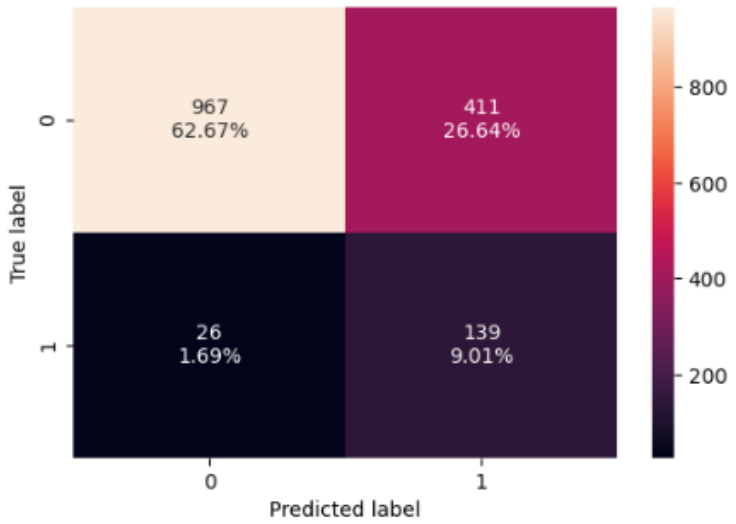

Figure 36: Confusion Matrix of Model

	Accuracy	Recall	Precision	F1
0	0.716785	0.842424	0.252727	0.388811

Figure 37: Model Performance

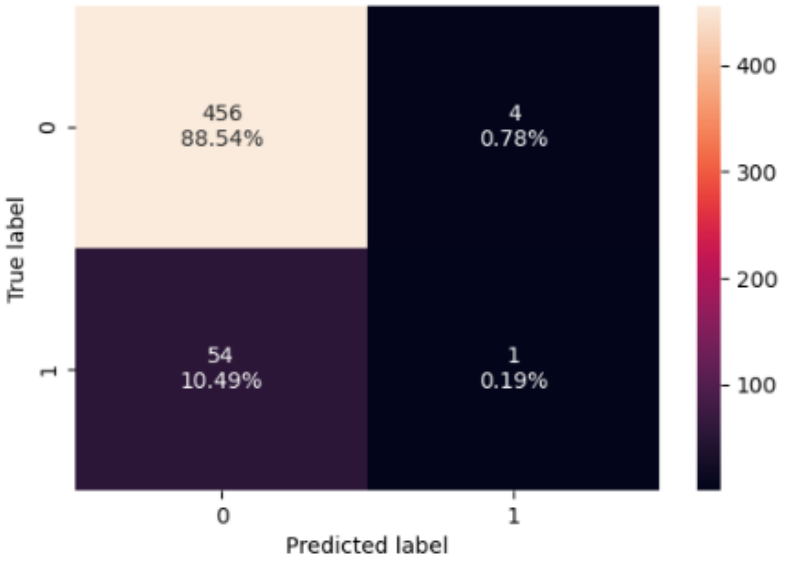

Figure 38: Confusion Matrix of Model

	Accuracy	Recall	Precision	F1
0	0.887379	0.018182	0.2	0.033333

Figure 39: Model Performance

23. Model Performance Improvement – Random Forest

```
Parameters used in the Random Forest Classifier:
bootstrap: True
ccp_alpha: 0.0
class_weight: balanced
criterion: gini
max_depth: 5
max_features: sqrt
max_leaf_nodes: None
max_samples: None
min_impurity_decrease: 0.0
min_samples_leaf: 6
min_samples_split: 2
min_weight_fraction_leaf: 0.0
monotonic_cst: None
n_estimators: 100
n_jobs: None
oob_score: False
random_state: 42
verbose: 0
warm_start: False
```

Figure 40: Model Parameters

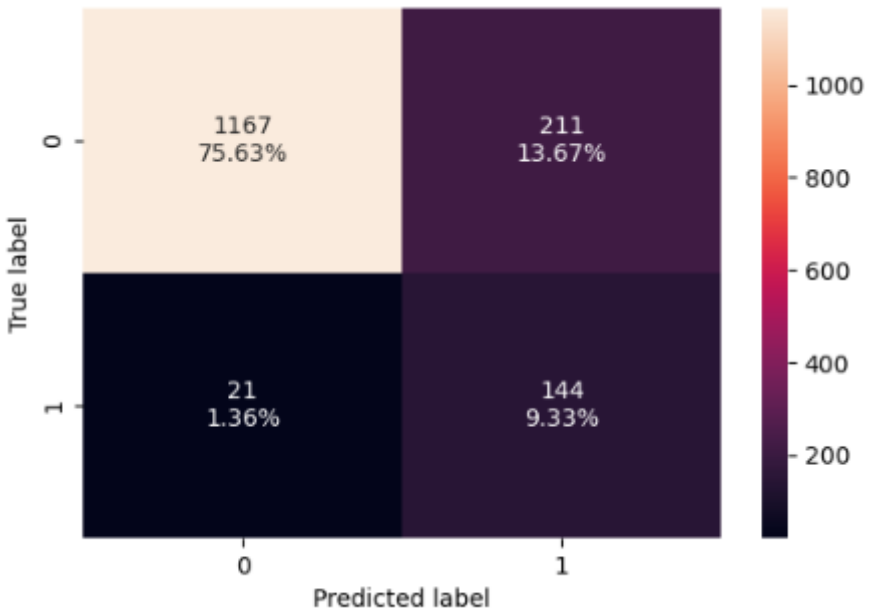

Figure 41: Confusion Matrix of Model

	Accuracy	Recall	Precision	F1
0	0.849644	0.872727	0.405634	0.553846

Figure 42: Model Performance

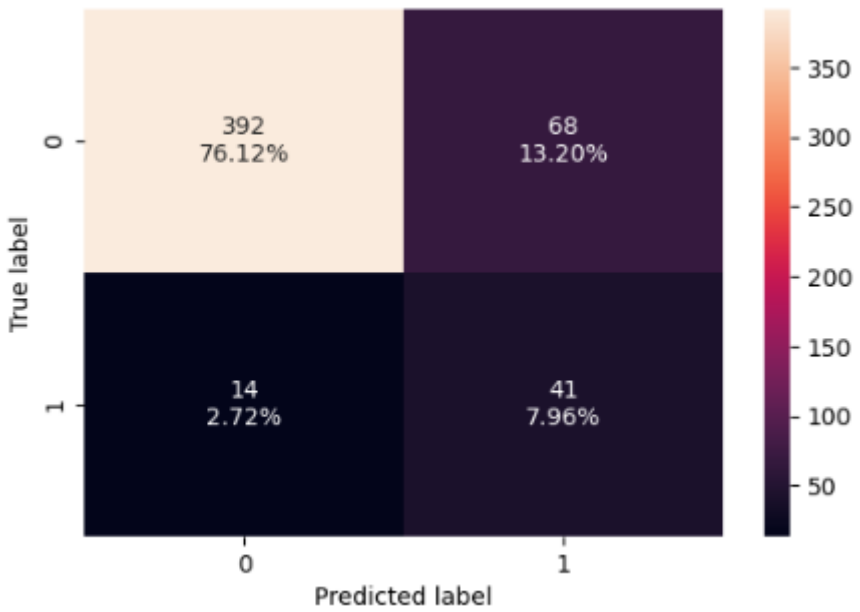

Figure 43: Confusion Matrix of Model

	Accuracy	Recall	Precision	F1
0	0.840777	0.745455	0.376147	0.5

Figure 44: Model Performance

24. Model Comparison and Final Model Selection

Training performance comparison:

	Logistic Regression	Tuned Logistic Regression	Random Forest	Tuned Random Forest
Accuracy	0.922229	0.716785	1.0	0.849644
Recall	0.460606	0.842424	1.0	0.872727
Precision	0.710280	0.252727	1.0	0.405634
F1	0.558824	0.388811	1.0	0.553846

Figure 45: Model Comparison on Training Data

Testing performance comparison:

	Logistic Regression	Tuned Logistic Regression	Random Forest	Tuned Random Forest
Accuracy	0.916505	0.887379	0.930097	0.840777
Recall	0.436364	0.018182	0.490909	0.745455
Precision	0.666667	0.200000	0.771429	0.376147
F1	0.527473	0.033333	0.600000	0.500000

Figure 46: Model Comparison on Test Data

Tuned Random Forest is the best model for this data

25. Important Features in Final Model

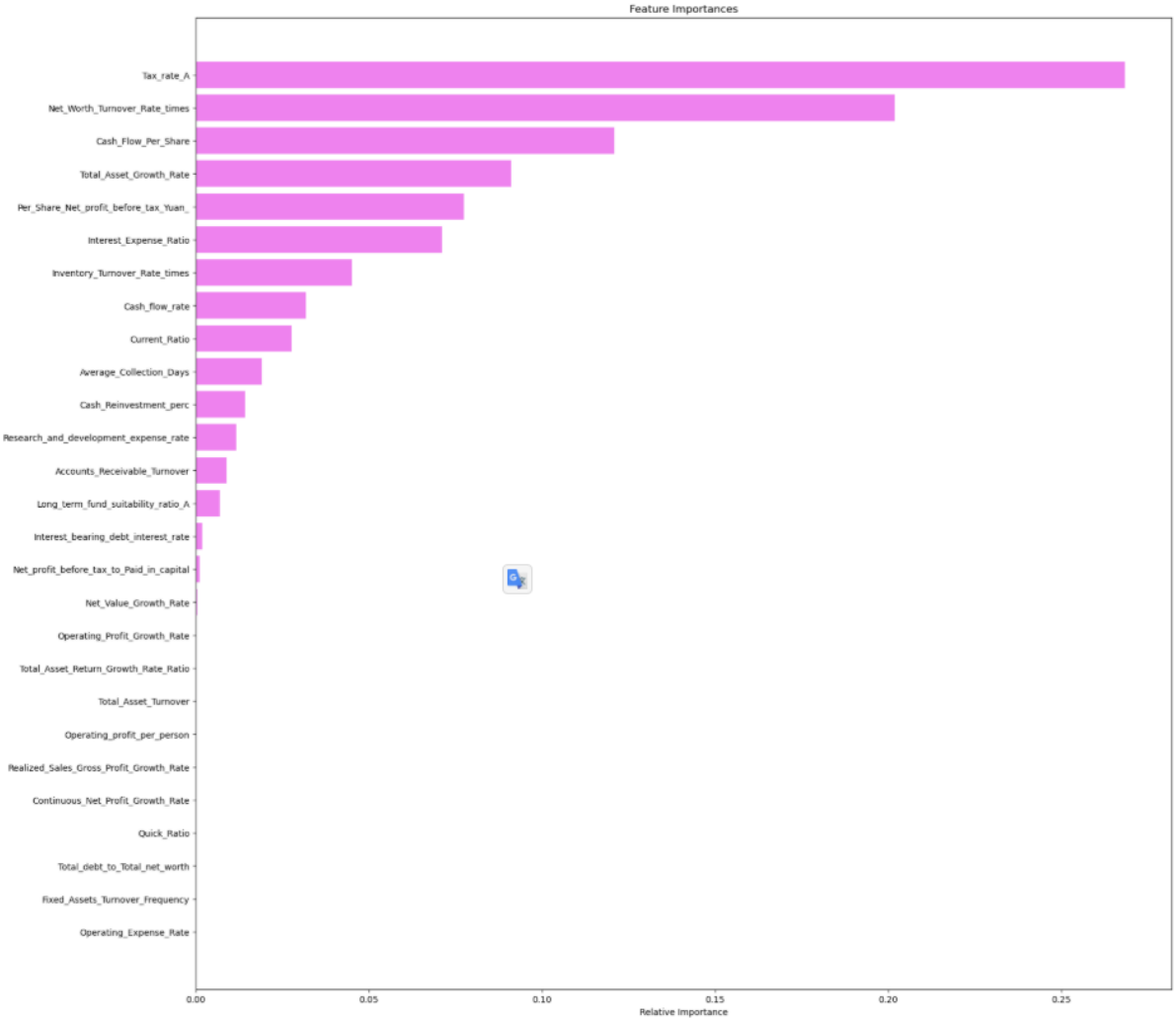

Figure 47: Important Features in Final Model

26. Insights and Recommendations

Insights

- Cash Flow Strength is a Critical Indicator of Default Risk Companies with low values in features like Cash_Flow_Per_Share and Cash_Flow_to_Liability are significantly more likely to default.
- Tax Rate and Equity Efficiency Reflect Financial Stress A high Tax_rate_A (Effective Tax Rate) and low Net_Worth_Turnover_Rate_times (Equity Turnover) are strong predictors of default.
- Leverage and Solvency Ratios Matter Ratios like Interest_bearing_debt_interest_rate and Total_debt_to_Total_net_worth are consistently higher in defaulting companies.
- Growth Metrics Alone Are Not Sufficient Features like Total_Asset_Growth_Rate or Operating_Profit_Growth_Rate were not among the top predictors.

Recommendations

- Prioritize Cash Flow Metrics in Credit Evaluation Incorporate cash flow–based ratios as primary indicators in the Financial Health Assessment Tool.
- Develop a Risk Scoring System Based on Top Predictors Build a transparent scoring model that weights features like Tax_rate_A, Cash_Flow_Per_Share, and Net_Worth_Turnover_Rate_times.
- Enable Dynamic Threshold Adjustment for Risk Appetite Allow users (e.g., investors or analysts) to adjust the classification threshold to balance recall and precision based on their tolerance for false positives.
- Monitor Leverage and Solvency Trends Over Time Integrate trend analysis for key ratios like Debt-to-Equity and Interest Coverage to detect deteriorating financial health early.
- Educate Clients on Interpreting Financial Red Flags Provide guidance within the tool to help users understand why a company is flagged.
- Retrain and Validate the Model Periodically Financial patterns evolve with market conditions. Regularly update the model with new fiscal data to maintain predictive accuracy and relevance.